

AWS 3-Tier Architecture Project

Submitted To: AWS LAB

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1. Project Overview

Project Title

Deployment of a Web Application Using AWS 3-Tier Architecture

Objective

The objective of this project is to deploy a scalable and secure web application using AWS cloud services following a 3-tier architecture model.

Description

A 3-tier architecture separates the application into three layers:

1. Presentation Layer (Web Tier)
2. Application Layer (Business Logic Tier)
3. Database Layer (Data Tier)

This architecture improves scalability, security, maintainability, and performance.

2. Architecture Design

Web Tier

- * Amazon EC2 Instance
- * Apache/Nginx Web Server
- * Hosts the frontend application

Application Tier

- * Amazon EC2 Instance
- * Application Server
- * Processes business logic

Database Tier

- * Amazon RDS (MySQL)
- * Stores application data

• Architecture Flow

User Browser



Web Server (EC2)



Application Server (EC2)



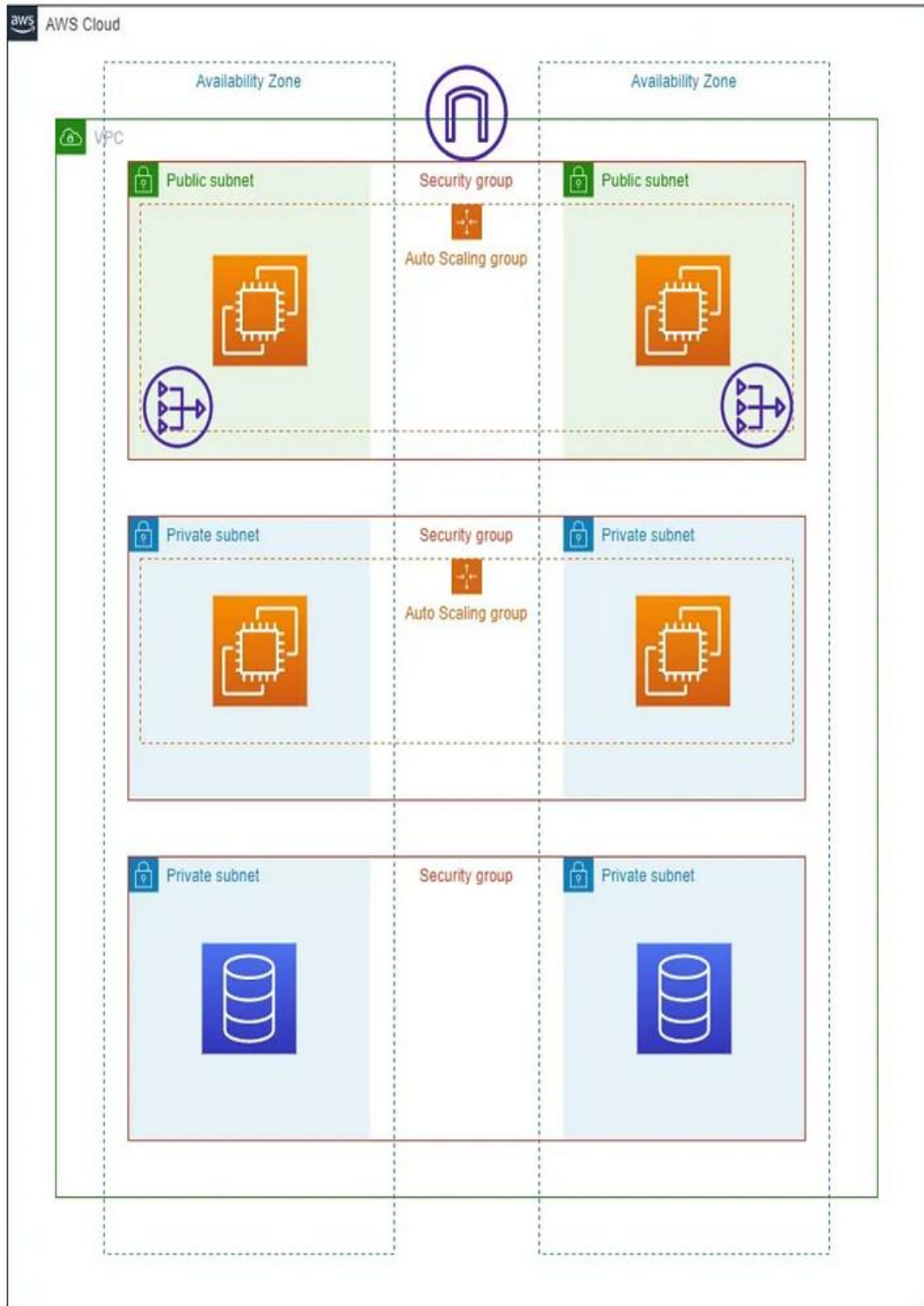
Amazon RDS (MySQL)

Security Groups:

* Internet → Web Tier (HTTP/HTTPS)

* Web Tier → App Tier

* App Tier → Database Tier

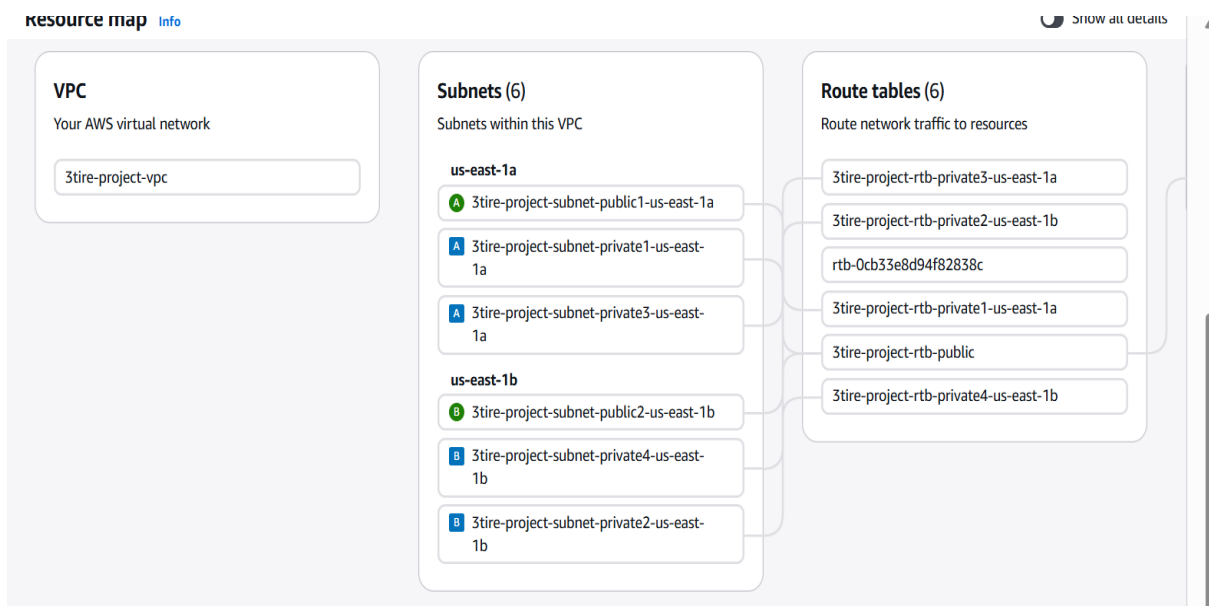


I am going to show you here how to build a 3 tier architecture in AWS

Part 1: Creating a VPC and Subnets

We need to create a VPC (Virtual private cloud) where this whole architecture will be housed in. We need 1 VPC, 2 public subnets and 4 private subnets under this VPC.

Log in to the AWS, go to the VPC section and click on “Create VPC”. There are two options here “VPC only” or “VPC and more”. We will be creating a VPC with multiple subnets, availability zones and security groups and more — so we will need to click “VPC and more”. You can use tag generation which will tag all of the resources under your VPC.



Click “create VPC” when you are ready. Once the VPC is created you can click “View VPC” to see all the resources that were created.

vpcs | VPC Console [x](#) [+](#) [Ask Gemini](#) [-](#) [x](#)

[←](#) [→](#) [↺](#) [↻](#) [https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs:](#) [☆](#) [🔍](#) [🌐](#) [🔔](#) [🔒](#) [⚙️](#) [United States \(N. Virginia\)](#) [naga \(454697937360\)](#) [naga](#)

[aws](#) [EC2](#)

[VPC](#) > Your VPCs

VPC dashboard [←](#)

[AWS Global View](#) [↗️](#)

[Filter by VPC](#) [▼](#)

▼ **Virtual private cloud**

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

▼ **Security**

Your VPCs

[VPCs](#) | [VPC encryption controls](#)

Your VPCs (1/2) [Info](#)

[Q](#) Find VPCs by attribute or tag

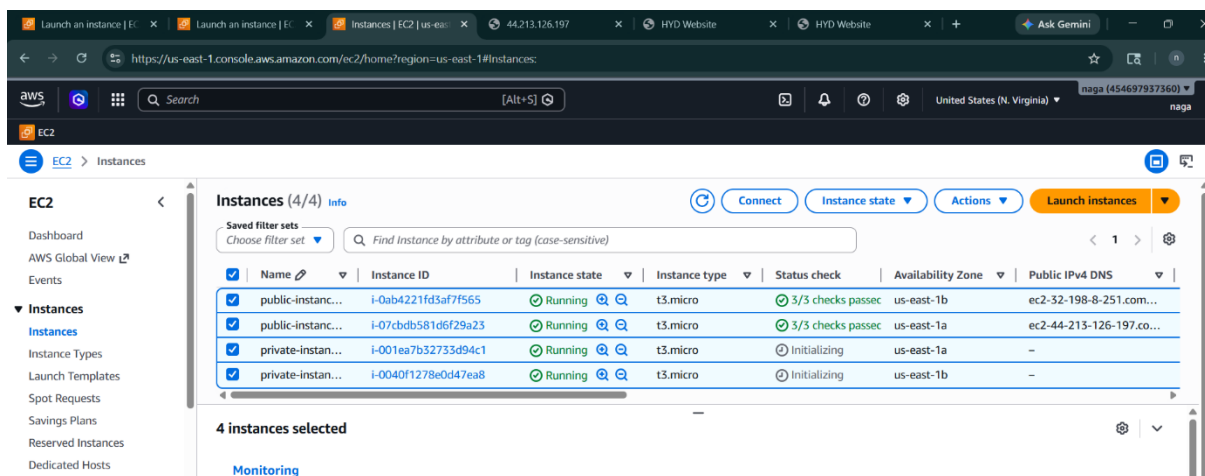
Last updated less than a minute ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
☐	-	vpc-016f6c2e0c14636b4	Available	-	-	Off	172
☒	3tire-project-vpc	vpc-0aee45a88b51ad40c	Available	-	-	Off	10.0

[vpc-0aee45a88b51ad40c / 3tire-project-vpc](#) [^](#)

Part 2: Create EC2 instances

Go to the EC2 console in AWS and click “Launch Instances”. You can name your server anything you want. Now go and create 4 ec2 instances. 2 public instances (public subnets 1,2), 2 private instances (private subnets 1,2) with different availability zones.



Once your instance is launched and running, you can select this instance and paste the public IPv4 address on your browser to check the custom webpages.

Public instance-1



Public instance -2



Welcome to the HYD Website

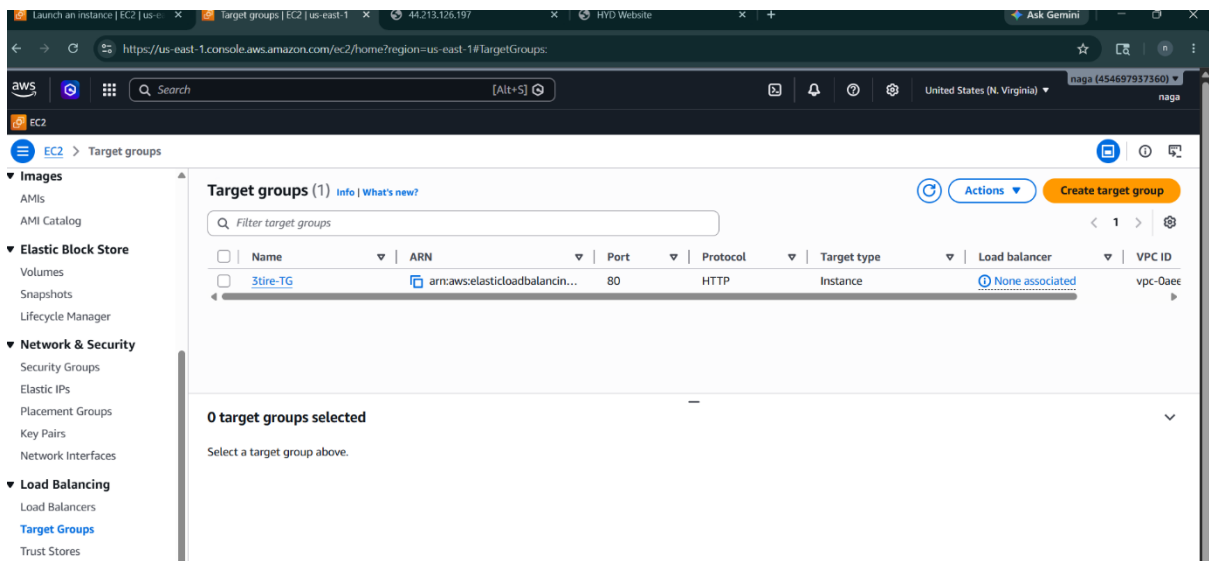
I hate HYD

Now we need to create target group and load balancer

First create a target group

1. give name (3tire-tg)

2. select 2 public instances, then click on include as pending below,

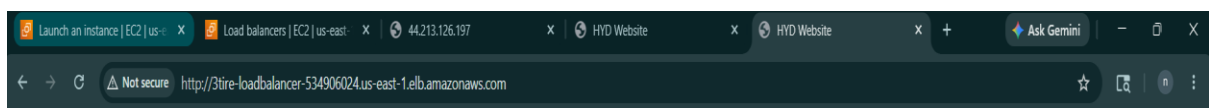
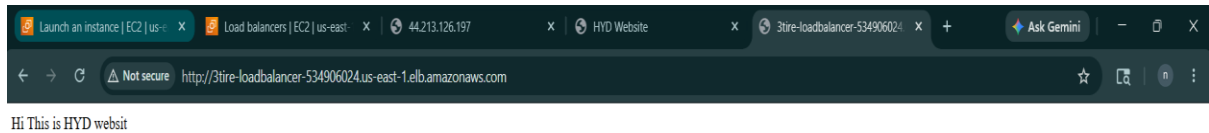


Now create application load balancer.

1. name (3tire-load balancer)
2. select created vpc, then availability zones.
3. select security group (3tire- sg).
4. select target groups (3tire-tg).
5. click on create load balancer

- After when it is available copy the DNS name URL and paste it on the new tab it will the information which u have entered in

webserver. Do refresh it will switch between two servers because application load balancer has round robin.

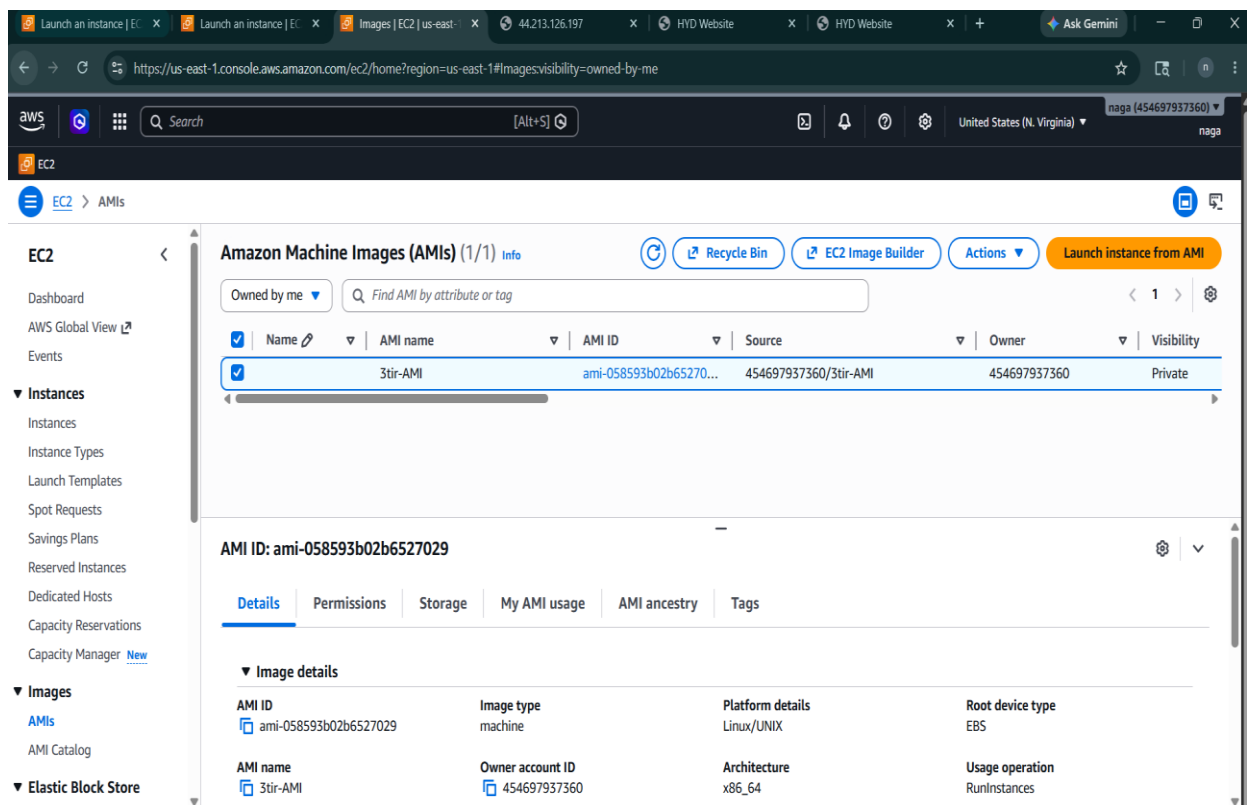


Welcome to the HYD Website

I hate HYD

- After this we need to create AMI. Go to public instance 1 click on actions select image and templates in that click on create image.

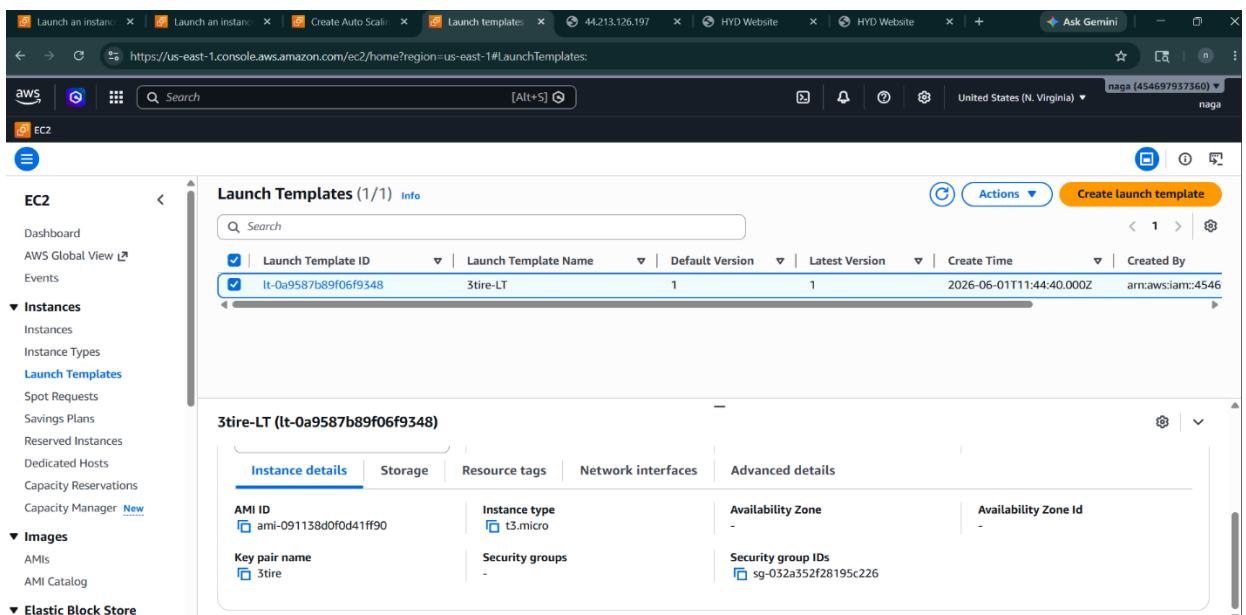
1. Enter image name
2. click on create image now go to AMI's
check whether created image is
available



Now create Auto scaling group to create
ASG we need to have launch template.
First create launch template then go and
create Auto Scaling Group

Launch template:

1. Name.
2. select AMI.
3. select keypair, instance type, security group.
4. click on create launch template.

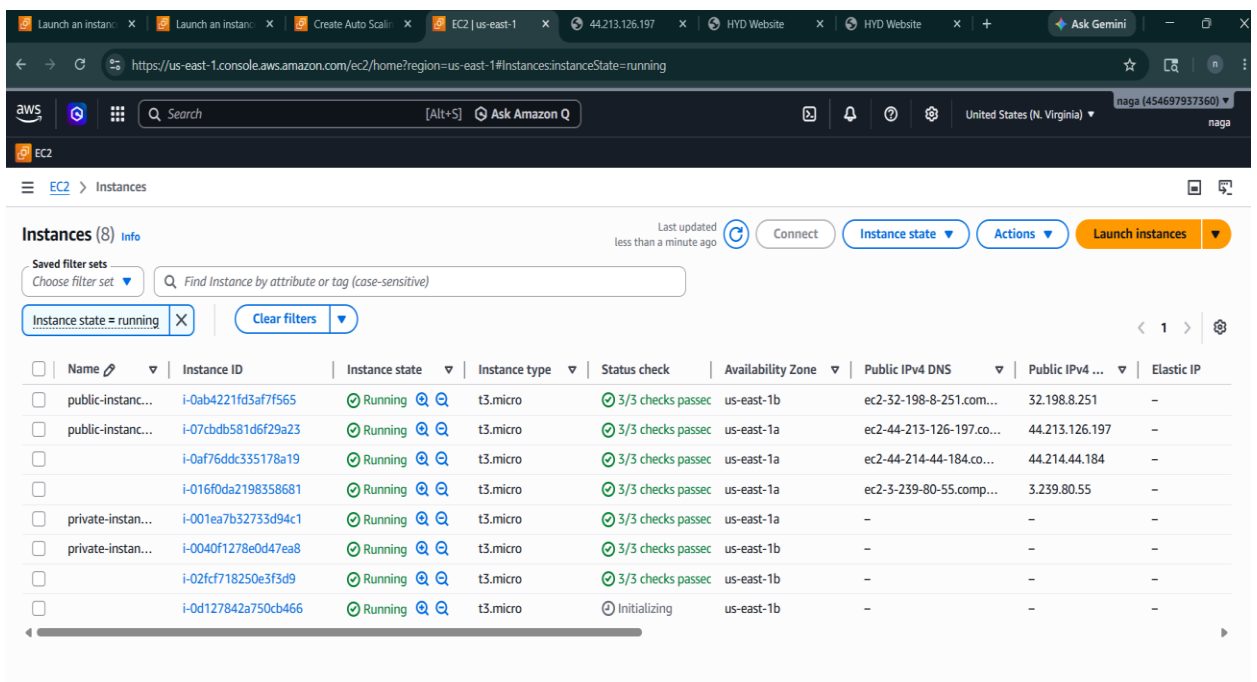


Auto Scaling Group:

1. Name.
2. select launch template (select created launch template).
3. select vpc, availability zones, subnets

4. select existing load balancer, target groups
5. select desired capacity as per your requirement I have taken 4, In target tracking scaling policy set target values (40) and instance warmup (300) as per your requirement.
6. click next, then click on create auto scaling group

Another 4 EC2 instances created by auto scaling group.



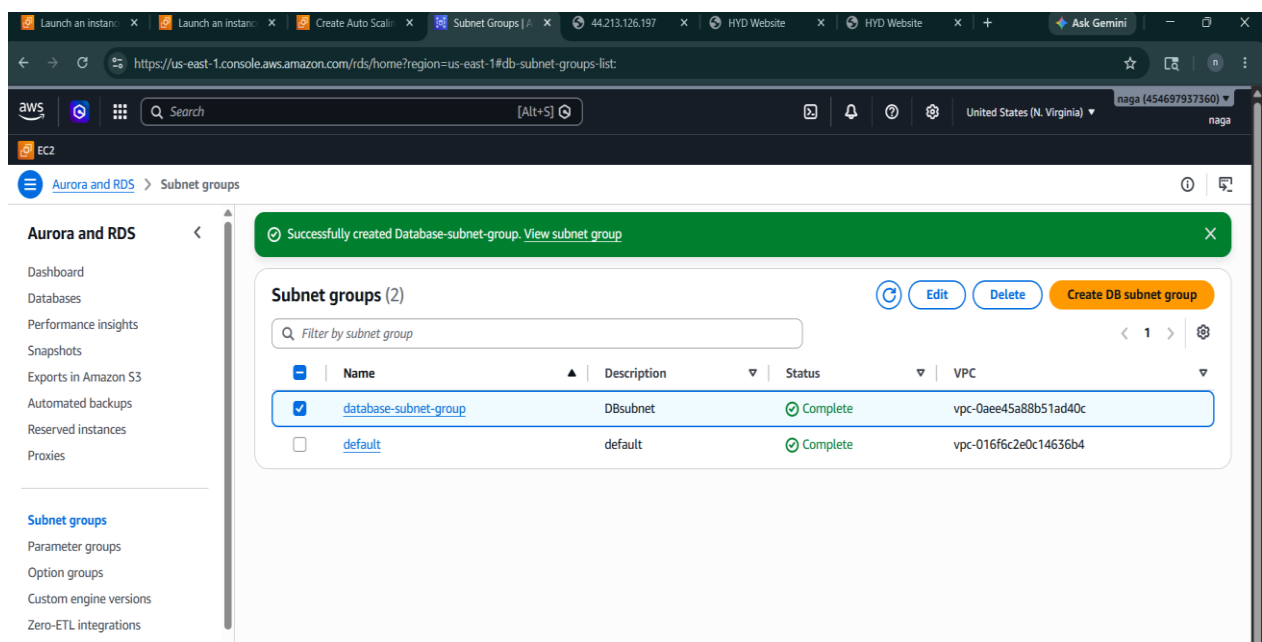
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	public-instanc...	i-0ab4221fd3af7f565	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-32-198-8-251.com...	32.198.8.251	-
☐	public-instanc...	i-07cddb581d6f29a23	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-213-126-197.co...	44.213.126.197	-
☐		i-0af76ddc335178a19	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-214-44-184.co...	44.214.44.184	-
☐		i-016f0da2198358681	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-239-80-55.comp...	3.239.80.55	-
☐	private-instanc...	i-001ea7b32733d94c1	Running	t3.micro	3/3 checks passed	us-east-1a	-	-	-
☐	private-instanc...	i-0040f1278e0d47ea8	Running	t3.micro	3/3 checks passed	us-east-1b	-	-	-
☐		i-02fcf718250e3f3d9	Running	t3.micro	3/3 checks passed	us-east-1b	-	-	-
☐		i-0d127842a750cb466	Initializing	t3.micro	Initializing	us-east-1b	-	-	-

Part 3: create RDS database

We need to create RDS database. Before creating it first create subnet groups.

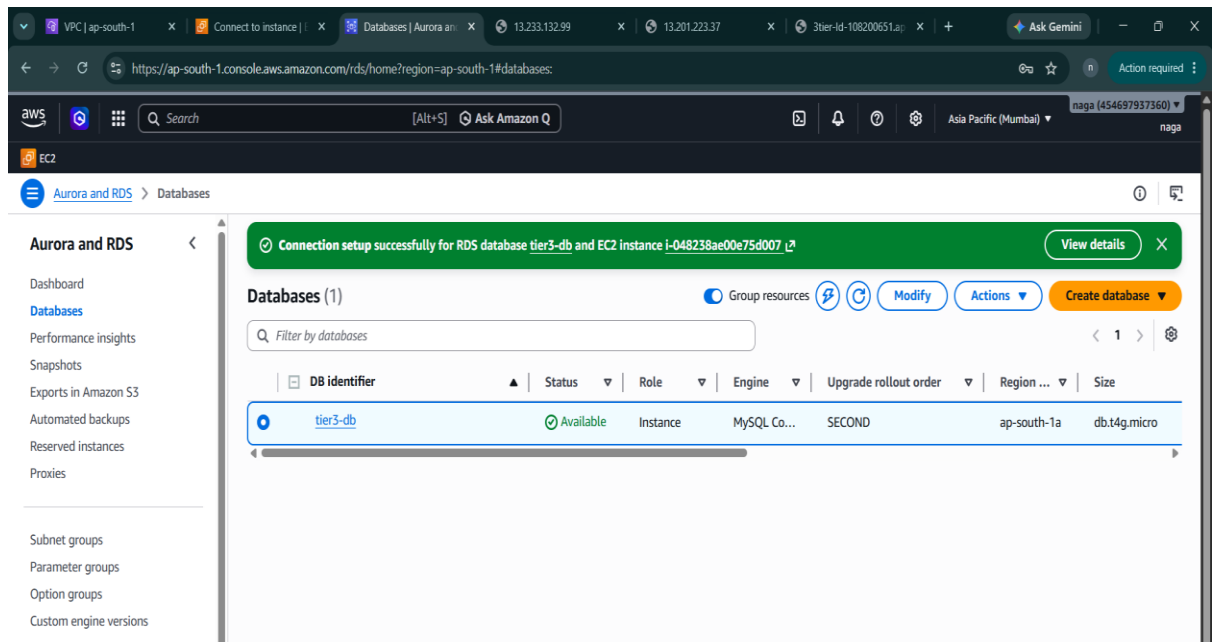
Subnet groups:

1. name (private subnet group),
description (allow)
2. select vpc
3. select availability zones, subnets
(private subnets 3,4).
4. click on create subnet groups.



RDS:

1. Click on create database, select standard create, select engine type as MySQL.
2. Select templates as production and select multi-AZ DB cluster.
3. Select on self-managed, give password and confirm the password.
4. Select memory optimized class.
5. In connectivity, click on Don't connect to the EC2 compute resource and select created vpc.
6. Select subnet group and give public access as yes.
7. Choose existing security group.
8. Go to VPC dashboard, click on VPC, click on actions, go to edit VPC settings and click on the enable DNS hostnames.
9. Create the database.



- Connect to Your Public Web Server (EC2)
- Switch over to the root user to gain administrator access:
- Install the MySQL Client on the Server
apt update -y
apt install -y mysql-client
- After installing we need to login into mysql
mysql -h endpoint url -u admin -p
- Then enter password
- Now you will be logged into mysql
- Once we successfully logged into the database and saw the mysql> prompt

1. Show databases
2. create database naga;
3. use naga;
4. create table

```
CREATE DATABASE IF NOT EXISTS  
myapp_db;  
USE myapp_db;
```

```
CREATE TABLE IF NOT EXISTS users (  
    id INT AUTO_INCREMENT PRIMARY  
KEY,  
    name VARCHAR(50),  
    age INT,  
    address VARCHAR(100),  
    sex VARCHAR(10)  
);
```

```
TRUNCATE TABLE users;  
INSERT INTO users (name, age,  
address, sex)
```

VALUES ('Naga vamsi', 25, 'Hyderabad, India', 'Male');

SELECT * FROM users;

```
root@ip-10-0-8-244: ~
mysql> show databases;
+-----+
| Database |
+-----+
| Naga      |
| information_schema |
| mysql     |
| performance_schema |
| sys       |
+-----+
5 rows in set (0.01 sec)

mysql> use Naga;
Database changed
mysql> CREATE TABLE IF NOT EXISTS users (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(50),
  -> age INT,
  -> address VARCHAR(100),
  -> sex VARCHAR(10)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> TRUNCATE TABLE users;
VALUES ('Your Name', 25, 'Hyderabad, India', 'Male');Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO users (name, age, address, sex)
  -> VALUES ('Naga Vamsi', 25, 'Hyderabad, India', 'Male');
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM users;
+-----+-----+-----+-----+-----+
| id | name      | age | address          | sex |
+-----+-----+-----+-----+-----+
| 1  | Naga Vamsi | 25  | Hyderabad, India | Male |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Result:

Successfully deployed a secure and scalable AWS 3-tier architecture application consisting of web, application, and database layers.

Conclusion:

The project successfully demonstrated the implementation of a 3-tier architecture on AWS using EC2, RDS, VPC, and Security Groups. The architecture provided better scalability, security, and separation of responsibilities among application components. This project enhanced practical knowledge of cloud infrastructure deployment and AWS networking concepts

THANK YOU